

Vision-based UAV Altitude Estimation using Deep Learning: A ResNet50 Approach on Nadir Images

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Abstract

This study proposes a vision-based deep learning approach for unmanned aerial vehicle (UAV) altitude estimation as an alternative to traditional methods such as GPS, barometric sensors, and laser altimeters, which are often sensitive to environmental disturbances. A large-scale dataset of 303,710 nadir images was collected using Mavic 2 Pro and Mavic 2 Zoom platforms under diverse weather, illumination, and terrain conditions. Each image was labeled with above-ground-level (AGL) altitude by integrating EXIF-based GPS altitude with a 30 m Digital Elevation Model (DEM) through coordinate transformation and terrain subtraction. A pre-trained ResNet50 model, originally designed for image classification, was reconfigured as a regression network and fine-tuned for 200 epochs using the Adam optimizer and mean squared error (MSE) loss. The proposed model achieved a mean absolute error (MAE) of 4.09 m in urban areas and 6.06 m in rural areas, with R^2 scores of 0.9981 and 0.9804, respectively. Comparative experiments with alternative CNN architectures show that the adapted ResNet50 provides favorable accuracy versus complexity trade-off. These results indicate that the proposed monocular, nadir-image-based framework can serve as a reliable complement or alternative to conventional altitude sensors in UAV operations.

Keywords: UAV Altitude Estimation; Deep Learning; Digital Elevation Model (DEM); Nadir Aerial Images; ResNet50 for Regression.

1. INTRODUCTION

Unmanned aerial vehicles (UAVs) have become integral to a wide range of critical missions, including reconnaissance, tracking, and mapping, particularly in challenging environmental conditions [1,2]. Their applications extend beyond military and civilian operations, finding widespread use in precision agriculture, search-and-rescue missions, and infrastructure inspection [3,4]. The ability of UAVs to autonomously navigate and execute missions with high precision is crucial for their effectiveness [1,5]. A fundamental requirement for such autonomy is accurate and reliable altitude estimation, which directly affects mission success and operational safety.

Traditional altitude measurement methods, such as the Global Positioning System (GPS) and barometric sensors, are often limited by external factors such as weather conditions and signal obstructions. GPS accuracy can degrade in areas where signals are weak or blocked, while barometric sensors are susceptible to atmospheric pressure fluctuations, leading to inconsistencies in altitude readings [6,7]. Laser altimeters provide high-precision measurements but are costly and may introduce additional payload and

complexity to UAV systems [8]. These limitations create a need for alternative altitude estimation approaches that are both accurate and resilient to environmental disturbances.

In recent years, vision-based methods leveraging onboard cameras have emerged as promising alternatives for UAV altitude estimation [9,10]. Unlike traditional sensors, vision-based approaches are less affected by signal loss or atmospheric conditions, making them suitable for diverse operational environments. Several studies have explored the potential of monocular depth estimation for UAV navigation, where deep learning models infer altitude from a single aerial image [11-13]. For example, Zhang et al. [14] successfully applied deep learning to monocular images for altitude estimation, demonstrating its feasibility. However, many of these studies utilized limited datasets or lacked robustness across varying environmental and flight conditions.

Deep learning, particularly convolutional neural networks (CNNs), has shown remarkable performance in image-based perception tasks, including depth estimation and UAV navigation [15-17]. CNNs can extract complex spatial patterns from aerial images, enabling accurate altitude

estimation from single-frame nadir (downward-facing) images. Recent advancements in deep learning have further improved real-time UAV perception capabilities. For instance, Zheng et al. [18] proposed a lightweight monocular depth estimation framework optimized for UAV navigation in indoor environments, achieving high accuracy with minimal computational requirements. Similarly, Piyakawanich and Phasukkit [11] combined clustering techniques with deep learning to enhance altitude estimation accuracy, demonstrating the potential of hybrid approaches. Additional studies focusing on monocular depth estimation in UAVs have introduced techniques such as edge-enhanced transformers or real-world self-supervision[13,19,20], further expanding the possibilities for robust, real-time UAV operations.

Despite these advancements, a gap remains in developing a vision-based altitude estimation model trained on large-scale nadir image datasets captured under diverse flight and environmental conditions. Existing studies have often focused on stereo vision or structure-from-motion techniques, which require additional computational resources and may not be suitable for real-time applications [21,22]. This study aims to address this gap by proposing a deep learning-based altitude estimation model trained on a large and diverse nadir image dataset, enabling a more robust and generalizable solution. Unlike prior studies, which often relied on small-scale datasets or stereo vision techniques, our approach leverages a large-scale nadir image dataset and a transfer learning-based CNN model (ResNet50) optimized for altitude regression. This enables robust and near real-time altitude estimation across diverse terrains and conditions.

To achieve this, a pre-trained CNN model has been fine-tuned for altitude regression using UAV flight data. The model's performance is evaluated in various real-world scenarios, highlighting its potential as an alternative or complement to traditional altitude sensors. As deep learning continues to drive innovations in autonomous systems [23], vision-based altitude estimation represents a promising direction for enhancing UAV navigation and mission planning capabilities.

The remainder of this paper is organized as follows: Section 2 (Materials and Methods) presents detailed information on dataset acquisition, preprocessing procedures, model architecture, and the training process. Section 3 (Results and Discussion) reports the experimental findings and discusses the performance of the proposed model in both urban and rural environments. Finally, Section 4 (Conclusions) summarizes the key outcomes and suggests potential directions for future research.

2. MATERIALS AND METHODS

This study aims to estimate altitude using deep learning techniques from unmanned aerial vehicle (UAV) images. The method consists of three main stages: data acquisition and preprocessing, model training and model testing.

2.1. Data Acquisition and Preprocessing

The key stages of the data preprocessing pipeline are summarized below:

- Data Acquisition via UAV flight
- Extracting GPS coordinates and altitude from EXIF metadata
- Estimating UAV ground clearance by subtracting DEM from EXIF altitude
- Performing coordinate transformations
- Scaling for different camera models
- Data augmentation (e.g., rotation, zoom, etc.)
- Adjusting the final image size and color space

2.1.1. Data Acquisition via UAV flight

In the first stage, UAV-captured NADIR (vertically downward-facing) images were processed for use in the deep neural network model. These images were chosen over oblique views to avoid complex geometric corrections, such as warping, thereby ensuring a consistent terrain view with reduced distortion and simplifying integration with digital elevation models (DEMs) for more accurate altitude estimation.

Images were acquired using Mavic 2 Pro and Mavic 2 Zoom UAVs equipped with GPS-enabled cameras at various altitudes and locations. UAV flights over the Cappadocia region, including Ürgüp, Turkey, yielded a diverse dataset covering both urban environments and rural landscapes—ranging from agricultural fields to natural rock formations. This variety supports robust generalization for altitude estimation across different environmental and topographical conditions.

2.1.2. Extracting GPS coordinates and altitude from EXIF metadata

GPS latitude, longitude and altitude data were extracted from the EXIF metadata using the Piexif and Exifread libraries [24].

2.1.3. Performing coordinate transformations and estimating UAV ground clearance by subtracting DEM from EXIF-based altitude data

A medium-resolution (30 meters GSD) Digital Elevation Model (DEM) file was used to obtain above ground level (AGL) altitude values [25]. The GPS coordinates (WGS84) have been converted to DEM's coordinate system (UTM Zone 36N) using the Pyproj library for a more accurate calculation. Due to this transformation, it was ensured that the coordinates for each data point were exactly matched and the most accurate ground surface height information was obtained through DEM. The conversion of GPS coordinates(EPSC:4326) to UTM coordinates(EPSC:32636) [26] is performed with the following equation:

$$(E, N) = \text{Transformer.from_crs}(\text{EPSG:4326}, \text{EPSG:32636}) \quad (1)$$

$$\text{.transform}(\text{lat}, \text{lon})$$

$$X = k_0 \cdot N \cdot (\lambda - \lambda_0) \cdot \cos(\phi), \quad (2)$$

$$Y = k_0 \cdot \left(N \cdot \tan(\phi) \cdot \frac{(\lambda - \lambda_0)^2}{2} \right)$$

Where;

λ : Longitude

λ_0 : Central Meridian

ϕ : Latitude

k_0 : Scale factor (usually 0.9996)

$$N: \frac{a}{\sqrt{1-e^2 \sin^2 \phi}}$$

The flight altitude above ground level (AGL) is computed by subtracting the terrain elevation from the GPS altitude recorded in the image's EXIF metadata:

$$H_{agl} = h_{GPS} - h_{terrain} \quad (3)$$

Where, h_{GPS} represents the altitude above mean sea level (AMSL) obtained from the EXIF metadata, and $h_{terrain}$ denotes the terrain elevation extracted from the DEM. Figure 1 visualizes the process of obtaining the altitude value through Digital Elevation Model.

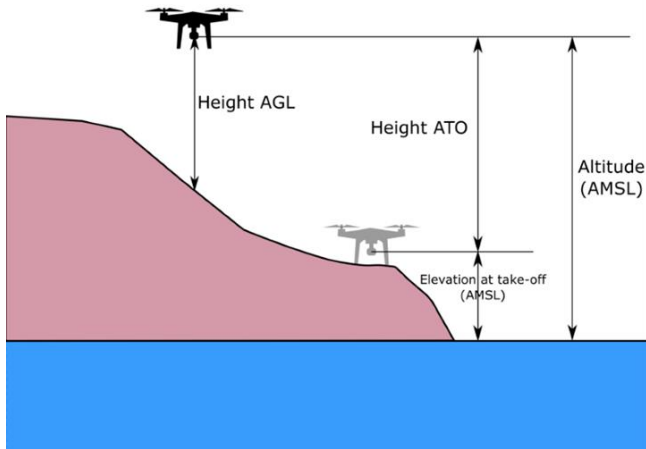


Figure 1. The process of obtaining the altitude value through DEM [25]

2.1.4. Scaling for different camera models

Given that the dataset comprises images captured by both the Mavic 2 Zoom and Mavic 2 Pro UAV platforms, it is imperative to harmonize the altitude estimates between these devices. In this study, the Mavic 2 Zoom is selected as the reference platform. Consequently, altitude values derived from the Mavic 2 Pro (L1D-20c) camera are adjusted to be compatible with those from the Mavic 2 Zoom. This correction compensates for differences in camera parameters-such as focal length, sensor size, and image resolution-that affect altitude estimation [27].

To convert the Mavic 2 Pro altitude estimates to the Mavic 2 Zoom equivalent, a scaling coefficient k is introduced. This

coefficient accounts for the inherent differences between the two camera systems and is defined as (see figure 2):

$$k = \frac{S_P}{f_P} \times \frac{f_Z}{S_Z} \times \frac{R_Z}{R_P} \approx 0.70 \quad (4)$$

Where;

S_P, f_P : Denote the sensor width and focal length for the Mavic 2 Pro, respectively

S_Z, f_Z : Denote the sensor width and focal length for the Mavic 2 Zoom

R_P, R_Z : Denote the image widths (in pixels) for the Mavic 2 Pro and Mavic 2 Zoom, respectively

In essence, this coefficient scales the altitude estimates from the Mavic 2 Pro so that they are directly comparable to those from the Mavic 2 Zoom. The adjusted flight altitude for Mavic 2 Pro images is then computed as:

$$h_{adjusted} = h_{flight} \times k \quad (5)$$

Where h_{flight} is the original flight altitude of the Mavic 2 Pro. Thus, by applying the scaling coefficient k , the final altitude estimate for Mavic 2 Pro images is standardized to the Mavic 2 Zoom equivalent. This standardization ensures uniformity in the training data and mitigates inconsistencies arising from hardware differences between the two UAV models.

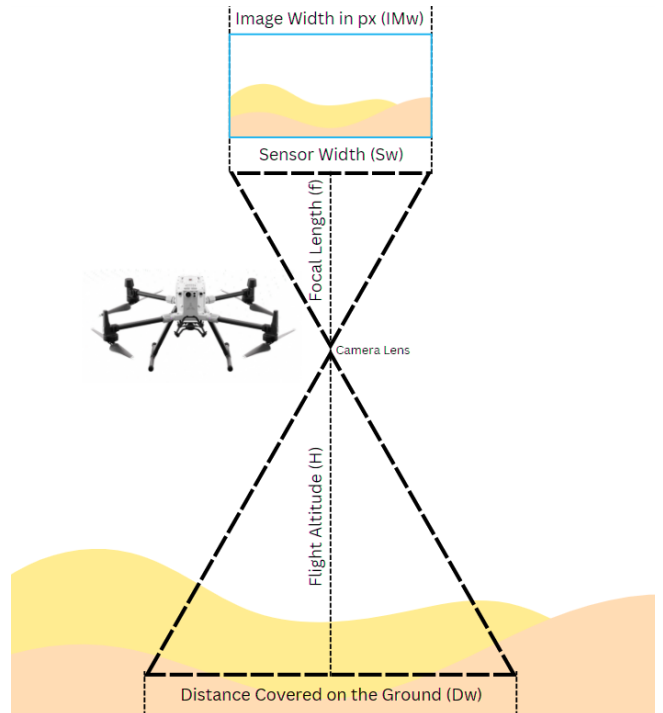


Figure 2. Ground distance covered based on camera parameters and altitude [17]

2.1.5. Data augmentation (e.g., rotation, zoom, etc.)

Images were rotated from 0 to 330 at 30-degree intervals to increase the model's ability to generalize to images at different angles (see Figure 4). To simulate images captured from various altitudes, zooming transformations with scaling

factors ranging from 0.4 to 0.95 were applied during the preprocessing stage. This approach generated data samples resembling images taken from different heights, enhancing the model's robustness to variations in altitude. After rotating and zooming, the images were cropped from the center to 1024×1024 pixels and resized to 512×512 pixels (see Figure 3). That pixel resolution was chosen to reduce computational cost during training while preserving sufficient detail. All images have been converted to RGB color space. Figure 5 shows a flowchart of the data augmentation steps.



Figure 3. The images were cropped from the center to 1024×1024 pixels and resized to 512×512 pixels for training.



Figure 4. Data augmentation strategies applied to nadir images, including rotational transformations at 30° intervals to improve model robustness across varying camera orientations

The dataset contains 303,710 images in total and covers different weather conditions (sunny, cloudy), different light conditions (sunrise, sunset, noon) and different terrain types (city, forest, farmland). This diversity aims to increase the generalization ability of the model. The calculated flight altitude was added to the EXIF data of the images under the GPS Altitude label. The processed images are saved in a specific output directory with file names that reflect the transformations applied.

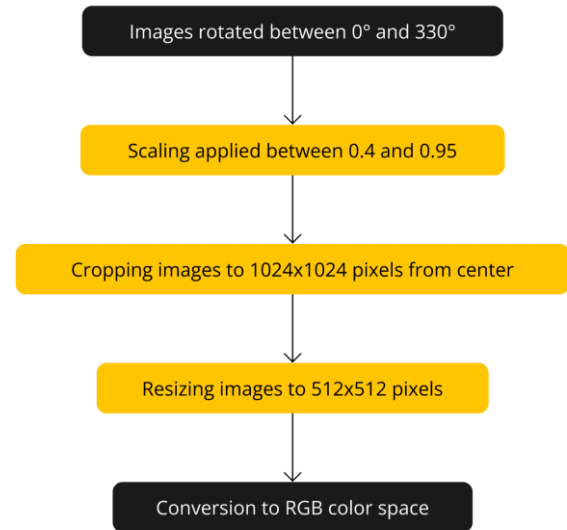


Figure 5. Flowchart of the data augmentation steps.

2.2. Model Architecture and Training Process

The file names and corresponding altitude values of the processed images are collected in a CSV file. The preprocessed dataset of 303,710 images was randomly split into 80% for training and 20% for validation. In addition, an independent test set of 1,781 nadir images was reserved and never used during training or validation. This test set consists of 762 urban and 1,019 rural samples, which are used for the quantitative evaluation reported in Table 1.

The image pixel values are scaled to the range $[0,1]$. However, no additional augmentations have been made here because the basic data augmentations have already been implemented.

The ResNet50 model [28], pretrained on the ImageNet dataset [29], was used as the base model for altitude estimation. It was selected due to its deep architecture and residual connections, which enhance feature extraction and mitigate vanishing gradients. Compared to alternatives such as VGG [30] and Inception [31], it offers an optimal balance between accuracy and computational efficiency. Its proven effectiveness in object recognition and depth estimation, combined with the availability of pretrained ImageNet weights, enables efficient transfer learning. Given these advantages, ResNet50 provides a robust and efficient framework for UAV-based altitude prediction. Before fixing ResNet50, we screened alternative backbones under the same split and preprocessing. Very lightweight models (MobileNetV2/V3 Small) reduced compute but degraded accuracy on our nadir imagery, while heavier or transformer-based models were excluded due to edge latency and memory targets as well as sensitivity to label noise from 30 m DEM labeling. We also ran short trials with VGG16 and a shallow convolutional autoencoder-based regressor; both underperformed on validation MAE, so we did not pursue them further as primary backbones. The latter is reported as "Custom CNN" in Table 1. ResNet50 therefore offered the best balance between accuracy and deployability in this setting.

To adapt the model for altitude estimation, the final classification layers of the model are removed and adapted for transfer learning. Global average pooling was used to reduce the size of the feature maps, then two fully connected layers with 1024 neurons were added, and ReLU was used as the activation function. A single neuron with a linear activation function has been added for altitude estimation. With this structure, the total number of parameters of the model is 26,736,513, of which 26,683,393 are trainable and 53,120 are non-trainable parameters (see Figure 6).

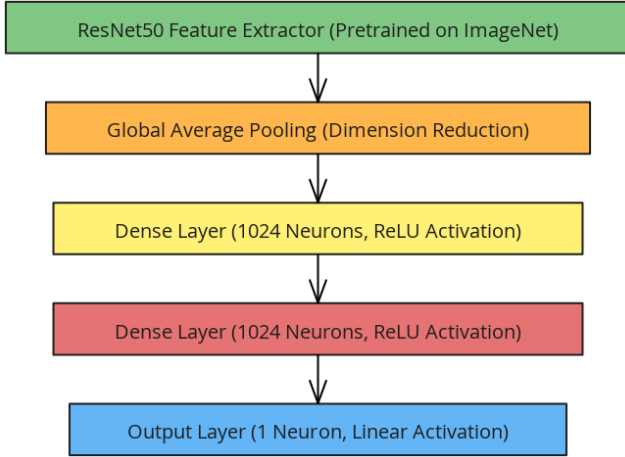


Figure 6. The architecture of the proposed model.

The model was compiled using the Adam optimization algorithm and the mean square error (MSE) loss function, with a learning rate of $\alpha = 5 \times 10^{-5}$. During training, the model was trained for 200 epochs with a batch size of 16. These hyperparameters were determined empirically through iterative experimentation, where initial values were selected based on commonly used configurations in similar deep learning tasks. Adjustments were made to balance convergence stability and computational efficiency while minimizing loss and preventing overfitting. Although no exhaustive grid search was conducted, multiple training runs were performed to identify an optimal combination that enhanced model performance in UAV-based altitude estimation.

To ensure robust model tracking and prevent unnecessary retraining, special callback functions were implemented to save model checkpoints at specific steps and at the end of each epoch. The mean square error (MSE) loss function, used to quantify prediction error, is defined as follows:

$$MSE = \frac{1}{n} \sum_{i=1}^n (y_i' - y_i)^2 \quad (6)$$

where y_i is the real altitude value, y_i' is the altitude value predicted by the model.

The training process was conducted using TensorFlow 2.5 on an NVIDIA RTX 4070 Super GPU with 12 GB VRAM and 64 GB system Ram. All experiments were performed on a Windows environment, ensuring efficient execution and

reproducibility. The implementation leveraged CUDA acceleration for optimized tensor operations, significantly reducing computational time. For inference, the trained model processes approximately 5.6 images per second at 512×512 resolution on the same GPU, which corresponds to about 0.18 s per image.

2.3. Performance Evaluation Metrics

In the final stage, the trained model was tested on new images and its performance was evaluated. GPS latitude, longitude and altitude information of each test image were extracted using the Exifread library. The distance of the locations where the test images were taken to the terrain was obtained using the DEM and coordinate transformations used earlier.

Test images were processed by reading them in RGB format, cropping them to 1024×1024 pixels from the center, and subsequently resizing them to 512×512 pixels using the nearest neighbor interpolation method. The pixel values are scaled to the range $[0,1]$. Processed test images were given to the model and altitude estimates were obtained. The difference between the estimated altitude value and the actual altitude value was found for each image. The mean absolute error (MAE) was calculated as:

$$MAE = \frac{1}{n} \sum_{i=1}^n |y_i' - y_i| \quad (7)$$

3. RESULTS AND DISCUSSION

The experimental results demonstrate that the developed model achieved high accuracy in both urban and rural environments, exhibiting strong predictive performance across diverse terrains. To validate the robustness of the proposed ResNet50-based architecture, we conducted a comparative analysis against alternative CNNs, including ResNet18, ResNet34, VGG16, MobileNetV2, MobileNetV3 Small, EfficientNetV2-B0, and a custom convolutional autoencoder (reported as "Custom CNN" in Table 1). For a fair comparison, all pre-trained models underwent the same transfer learning process: classification layers were replaced with a regression head to predict continuous altitude values. As detailed in Table 1, lightweight models like MobileNetV3 Small and the Custom CNN showed poor generalization (Rural R^2 : -0.87 and -9.98, respectively), likely due to the loss of spatial fidelity in separable convolutions or insufficient inductive bias. In contrast, the proposed residual-based model significantly outperformed even standard ResNet variants (Urban MAE: 4.09 m vs. 22.20 m for ResNet34). These results indicate that deeper model capacity and the chosen training configuration are critical for precise altitude regression, rather than the mere presence of residual connections. The performance gap between ResNet50 and the shallower residual variant ResNet34 under identical settings supports this interpretation.

The model's test performance is summarized in Table 1.

Table 1. Quantitative comparison of different CNN architectures using Transfer Learning on the urban and rural test subsets.

Model Architecture	Urban Areas (762 images)				Rural Areas (1019 images)			
	MAE	MSE	RMSE	R ²	MAE	MSE	RMSE	R ²
Proposed Model (ResNet50)	4.09	32.08	5.67	0.99	6.06	75.00	8.66	0.98
EfficientNetV2-B0	14.90	326.64	18.07	0.98	32.22	1170.56	34.21	0.82
ResNet34	22.20	880.64	29.68	0.95	14.24	351.09	18.74	0.95
ResNet18	27.70	1243.61	35.26	0.93	17.69	536.61	23.16	0.92
VGG16	21.25	748.53	27.36	0.96	29.29	1227.81	35.04	0.81
MobileNetV2	52.40	3389.69	58.22	0.80	62.87	4486.88	66.98	0.31
Custom CNN	67.26	8665.74	93.09	0.48	207.14	70982.15	266.42	-9.98
MobileNetV3 Small	99.54	11450.28	107.01	0.32	106.00	12095.24	109.98	-0.87

3.1. Urban Areas

In urban areas, the model demonstrated outstanding performance, achieving an MAE of 4.0925 meters, an MSE of 32.0767, an RMSE of 5.6636 meters, and an R² score of 0.9981. These metrics indicate near-perfect predictive accuracy, which can be attributed to the well-defined structural elements and consistent visual patterns inherent to urban environments. Figures 7 and 8 visually depict the model's performance in urban settings, clearly illustrating the close alignment between the predicted and actual altitude values.

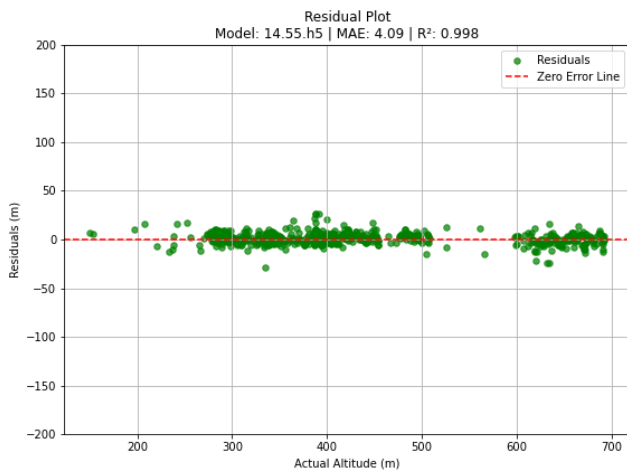


Figure 7. Residuals are tightly distributed around the zero-error line, reflecting a balanced error distribution with minimal deviations.

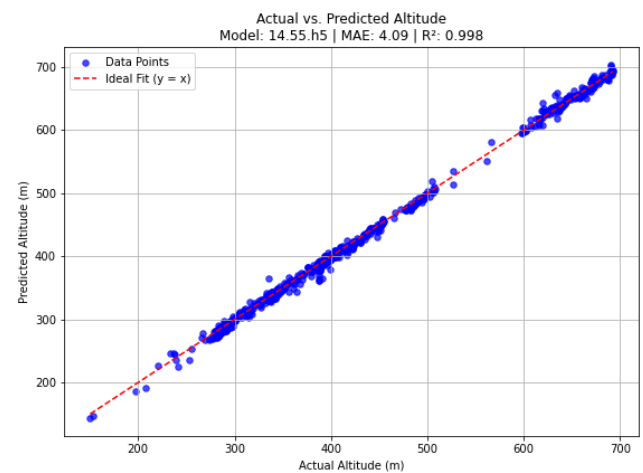


Figure 8. The alignment of data points along the ideal fit line ($y = x$) confirms the model's reliability in predicting altitude values in urban environments.

3.2. Rural Areas

In rural settings, where terrain features are more varied and less structured, the model maintained satisfactory performance, with an MAE of 6.0569 meters, an MSE of 75.0018, an RMSE of 8.6604 meters, and an R² score of 0.9804. However, the performance in these areas was slightly limited because ground truth AGL labels were derived using a DEM with 30 m GSD, which likely introduced label noise on rugged terrain. Given the highly rugged and uneven nature of the rural terrain, a higher resolution DEM could have provided more precise terrain data, thereby improving the model's predictive accuracy. Figures 9 and 10 illustrate the model's performance in rural areas, highlighting both the overall efficacy of the approach and the challenges introduced by the lower resolution DEM in capturing fine-scale topographic variations.

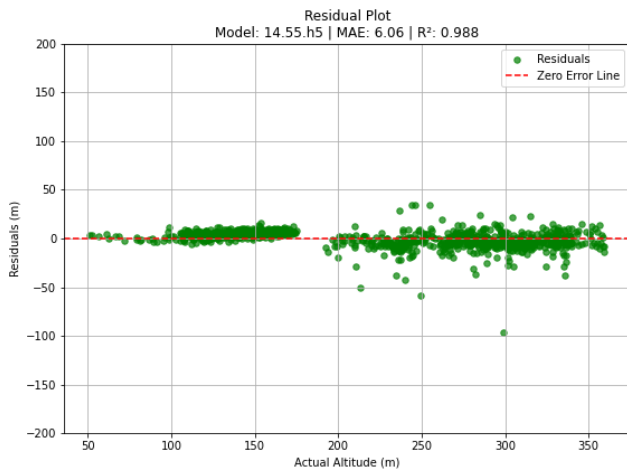


Figure 9. Residuals display a wider spread compared to urban datasets but remain relatively balanced around the zero-error line.

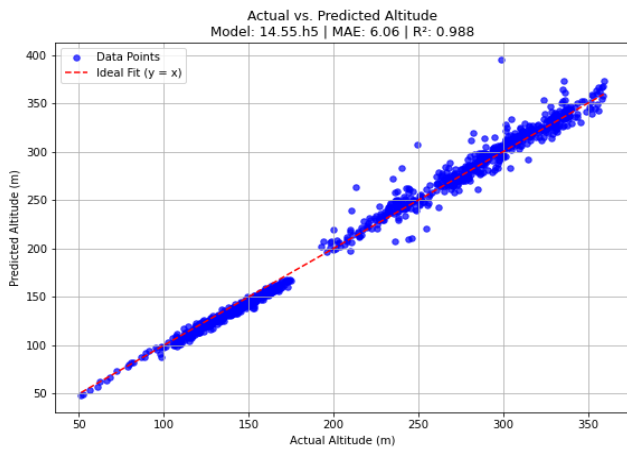


Figure 10. Data points are closely aligned with the ideal fit line, though some deviations are noticeable, particularly at higher altitudes.

3.3. Comparison with Existing Literature

Previous studies have explored various methodologies for UAV altitude estimation, each employing different approaches and techniques to address the challenges posed by varying environmental and operational conditions. For instance, Zhang et al. [14] integrated dense mapping techniques into stereo visual-SLAM systems, focusing primarily on reconstructing detailed environmental maps to support autonomous navigation and positioning. While their methodology provided robust spatial context, the approach involved higher computational complexity and reliance on stereo image processing, which may limit real-time applicability in resource-constrained UAV operations.

Similarly, Navarro et al. [21] utilized dense-direct visual-SLAM approaches designed for precise localization in close-range inspection tasks. Although these methods achieved impressive precision through stereo vision and detailed environmental modeling, their computational demands and hardware complexity are relatively high. Such factors could restrict their feasibility for deployment in standard UAV missions that require quick and scalable solutions.

Other researchers, including Piyakawanich and Phasukkit [11], introduced hybrid methodologies combining clustering and deep learning to enhance altitude estimation accuracy. These hybrid approaches generally leverage complex preprocessing steps and model integrations, potentially adding to computational overhead and processing latency.

Zheng et al. [18], on the other hand, proposed lightweight monocular depth estimation frameworks explicitly optimized for indoor UAV navigation scenarios. Their emphasis on computational efficiency and low resource usage aligns closely with operational constraints typical of small-scale UAVs. However, these methods are predominantly tailored for indoor environments, thus their adaptability and effectiveness in outdoor, more complex terrains could be limited.

In contrast to these methodologies, we present a monocular vision-based deep learning approach using a pre-trained ResNet50 for altitude estimation from single nadir UAV images. The main advantages are simplicity, computational tractability, and suitability for outdoor operations in both urban and rural settings. Although the ResNet backbone is heavier than MobileNet variants, our comparisons indicate a practically useful accuracy-latency trade-off, as lightweight models did not provide reliable estimates in our tests. On the reference desktop GPU, the trained network processes approximately 5.6 images per second at 512×512 resolution. This throughput is sufficient for slow surveying or hovering scenarios and indicates that near real-time inference is feasible under moderate flight speeds. However, further optimization and model compression would be required for high-speed autonomous flight or deployment on embedded platforms such as NVIDIA Jetson. We did not perform any on-device benchmarks in this study; our deployment discussion is based solely on desktop-GPU throughput and model size.

The current model can be exported to ONNX and integrated into embedded GPU runtimes, which we consider an important direction for future work.

However, this study is limited to the specific geography of the Cappadocia region. Future work will focus on expanding the dataset to diverse biomes (e.g., dense forests, maritime environments) to enhance global generalizability. Lighter backbones reduced compute but degraded accuracy in our data, so we keep this trade-off. When visual confidence drops due to blur or low light, the pipeline can slow or hover, and simple fusion with barometric readings improves robustness.

4. CONCLUSIONS

The high accuracy observed in both urban and rural datasets demonstrates the adaptability and robustness of the proposed model for UAV altitude estimation. Nevertheless, slightly lower performance observed in rural regions highlights specific challenges such as dense vegetation, uneven terrain, and sparse visual references. Particularly, the use of a 30-meter GSD DEM in rugged rural areas likely led to increased prediction errors due to its coarse resolution, which limits

capturing fine-scale topographic details. Additionally, occasional deviations at higher altitudes suggest that environmental conditions-such as variable lighting, diverse surface reflectance, and atmospheric effects-may also influence estimation accuracy.

In this study, we did not yet include Grad-CAM or other explainability maps, and the model is evaluated purely based on its predictive performance. In future work, we plan to perform additional flights following more systematic, grid-based trajectories and to generate a geospatially continuous Grad-CAM layer over the study area, so that both altitude predictions and their explanatory heatmaps can be analyzed as a single map-based product in safety-critical scenarios.

Future research could significantly enhance model capabilities by incorporating LiDAR-based labels, reducing inaccuracies arising from low-resolution DEMs, and thereby improving model training and evaluation reliability. Expanding dataset diversity to include nighttime imagery, varied weather scenarios (e.g., fog, rain, intense sunlight), and broader terrain types (such as forested, coastal, and mountainous regions) is another critical step. This expansion would strengthen the model's ability to generalize across different operational environments. Moreover, integrating supplementary sensor data, such as inertial measurement unit (IMU) readings, could provide valuable contextual insights, especially in environments with limited or ambiguous visual cues.

Adopting higher-resolution DEMs, particularly in rugged rural landscapes, may also significantly enhance estimation accuracy by capturing more detailed terrain variations. Additionally, making both the dataset and the trained model publicly available would foster collaboration within the research community, facilitating collective improvements and inspiring new applications.

In summary, this study demonstrates the feasibility and potential advantages of a vision-based deep learning approach for UAV altitude estimation using nadir imagery. Its strong performance in urban settings and adequate performance in rural areas indicate that the proposed model is a promising complement or alternative to traditional GPS and barometric sensors. Despite the encouraging outcomes, limitations associated with DEM resolution and rural landscape complexity highlight the need for further refinement. Continued efforts such as integrating higher-resolution DEMs, additional sensor data, and LiDAR-based labeling will enhance precision. While current findings provide an initial proof of concept, further development is essential before confidently deploying the approach in critical UAV missions. With ongoing improvements, expanded datasets, and collaborative open-source initiatives, this method has significant potential to advance UAV-based altitude estimation and promote progress in autonomous aerial technologies.

Data Availability Statement

The data presented in this study are available from the author upon reasonable request.

Abbreviations

The following abbreviations are used in this manuscript:

UAV	:	Unmanned Aerial Vehicle
DEM	:	Digital Elevation Model
RGB	:	Red, Green, Blue (color channels)
AGL	:	Above Ground Level
GPS	:	Global Positioning System
MAE	:	Mean Absolute Error
MSE	:	Mean Squared Error
RMSE	:	Root Mean Squared Error
R ²	:	Coefficient of Determination
CNN	:	Convolutional Neural Network

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